Dawar Imam

Computer Vision Engineer

Data Science researcher and engineer specializing in Deep Learning, Computer Vision, and LLM-based applications, with a passion for automating, optimizing, and deploying scalable AI solutions.

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WORK EXPERIENCE

Machine Learning Engineer
Xeven Solutions (Pvt) Ltd.

12/2024 - 07/2025 — *Plot 15, Civic Centre Block D 2 Phase 1 Johar Town, Lahore*

Xeven is a leading provider for delivering innovative, AI-driven, scalable, data-centric, and cloud-based digital solutions for businesses.

Achievements/Tasks

- Designed and automated the *Candidate Monitoring Module* for AI-powered online interviews, integrating *pose estimation*, *face verification* , and *gaze detection* using *WebSocket* (*Python backend*) and in-browser AI modules with *JavaScript* and *ONNX* .
- Leveraged *RAG pipelines*, *LangChain*, and *LangGraph* to automate recruitment workflows, utilizing *vector stores*, *embedding models*, and managing both *SQL (MSSQL)* and *NoSQL (Redis)* databases.
- Developed a *proof-of-concept (POC)* for *document forgery detection* using *LLMs* and *LangChain*, exploring AI-based alternatives to traditional *computer vision* methods.

Research Assistant
Intelligent Machines Lab, ITU

06/2023 - 11/2024 — *Arfa Karim Tower, Ferozpur Rd, Lahore, Pakistan*

Research lab focused on developing AI systems related to Domain Adaptation, 3D Scene Reconstruction, and Satellite Imagery

Achievements/Tasks

- Submitted a research article to *CVPR 2025*, proposing a regularization strategy in *Gaussian Splatting* using motion information for *Dynamic Scene Reconstruction* .
- Gained experience in *Gaussian Splatting-based methods* for *3D and 4D Scene Reconstruction*, working with *multiview stereo datasets* and *Structure-from-Motion (SfM)* based models.
- *Worked with Gaussian Splatting visualization tools* such as *SIBR Viewer*, *SfM* tools like *COLMAP*, and applied knowledge of *3D geometry* and *camera calibration* .
- Contributed to a research project on S*ynthetic Dataset Generation* for evaluating advanced scene reconstruction methods, using *Unity Game Engine* , *Blender* for 3D modeling, and setting up *PyTorch GPU environments* for deep learning experiments.

EDUCATION

Bachelors of Science in Computer Science
Information Technology University

06/2020 - 06/2024 — *Lahore, Pakistan*

Courses

- Deep Learning,
- Classical Computer Vision,
- Machine Leanring,
- Generative AI (Audit),

SKILLS

Python · C/C++ · Java · PyTorch · Langchain/Langraph · LLMs · Transformers · HuggingFace · ONNX · Git/GitHub · Docker · FastAPI · SQL · Conda · Unity (3D)

PERSONAL PROJECTS

Image Rectification and Panoramic Stitching using SIFT Detector

- A classical computer vision problem where we used the Scale Invariant Feature Transform (SIFT) detector for highlighting essential features for panoramic image stitching.

Face Recognition Classifier with Dimensionality Reduction using PCA

- Used Principle Component Analysis (PCA) for reducing data dimensionality for efficient training of CNN based classifier over a facial image dataset.

ACITIVITIES

Winter Game Jam (MindStorm Studio) (11/2021 - 12/2021)

Participated in the Winter Game Jam of Mind-Storm Studios for developing a hyper-casual 3D game in Unity.

LANGUAGES

Urdu — *Native or Bilingual Proficiency*

English — *Full Professional Proficiency*

INTERESTS

Pets · Sketching · Music